

Catalyst Screening through Quantum Chemical Calculations and Microkinetic Modeling: Hydrolysis of Carbon Dioxide¹

Renhu Ma^a, George F. Schuette^b, and Linda J. Broadbelt^{a, *}

^a Department of Chemical and Biological Engineering, Northwestern University, Evanston, IL 60208, USA

^b Phillips 66 Company, Bartlesville, OK, USA

*e-mail: broadbelt@northwestern.edu

Received May 11, 2017

Abstract—Quantum chemical calculations are emerging as an effective way to screen catalysts for particular applications. In this contribution, we demonstrate the power of density functional theory to study CO₂ hydrolysis by six carbonic anhydrase mimics, evaluating thermodynamic and kinetic parameters at the mechanistic level. A microkinetic model was then built based on the kinetics and thermodynamics calculated from first principles. The intrinsic reaction rate constant was calculated from the results of the microkinetic model and compared with experimental data. Overall, the rate constants were in good agreement with experimental values, except for zinc-tri and complex **b**, which were overestimated. This was ascribed to their ineffective complexation with Zn²⁺. How the reaction rate constants vary with time was also investigated. From 0 to 12 ms, the rate constants of complexes **a** and **d** decreased to 50 and 67% of their initial values, respectively; the rate constants of complexes **b** and **f2** were almost invariant with time; the rate constant of complex **f1** showed an unusual double sigmoidal shape. The p*K*_a values of these six carbonic anhydrase mimics as well as three additional mimics were calculated. A correlation between p*K*_a values and the binding free energy of OH[−] was obtained by fitting data from five zinc(II) aza-macrocyclic complexes. The reaction rate constants were found to increase linearly with the p*K*_a value, indicating CO₂ adsorption is the rate-limiting step.

Keywords: carbon dioxide hydrolysis, quantum chemical calculations, microkinetic modeling, zinc catalysts, kinetic analysis

DOI: 10.1134/S0040579517060124

INTRODUCTION

Greenhouse gases, particularly carbon dioxide from fossil fuel combustion, affect global climate by absorbing energy in the lower atmosphere and re-emitting it [1]. The amount of CO₂ in the atmosphere has been increasing significantly over the past 150 years—from a steady-level of 280 ppm before the industrial revolution to a current level of over 400 ppm [2]. The strong correlation between the level of CO₂ and global temperature urges us to develop rapid methods to capture and store CO₂ [3]. In industry, CO₂ adsorption is often achieved by converting CO₂ gas to carbonic acid in a liquid phase and subsequent removal of the liquid fraction [4]. The rate of CO₂ dissolution in water is limited by the rate of CO₂ hydrolysis, while the CO₂ capture capacity is determined by the buffering capacity. Adding catalysts to the solution would speed up the rate of bicarbonate formation and

thus improve the effectiveness of industrial separation processes [5]. In nature, carbonic anhydrase (CA) catalyzes the reversible hydrolysis of carbon dioxide with formidable efficiency. With a rate constant of 10⁶ M^{−1} s^{−1} HCA II (human carbonic anhydrase II) is the most efficient isoform, exhibiting activity that approaches the diffusion limit. The X-ray crystal structures of HCA II show that the active site includes a zinc atom tetrahedrally coordinating to three histidines (His-94, His-96, and His-119) and a catalytic water molecule (Wat-263), a hydrophobic pocket (Leu-198, Trp-209, Val-143, Thr-199, Val-207, and Val-121) for CO₂ binding, and a hydrophilic half (Tyr-7, Asn-62, His-64, Asn-67, Thr-199, and Thr-200) for containing water molecules and providing a hydrogen bonding network for proton shuttling during the last step of the catalysis [6, 7].

While enzymes are highly active and selective, they may not be sufficiently robust to withstand conditions relevant to industrial CO₂ capture. Thus, previous work has focused on developing small molecule mim-

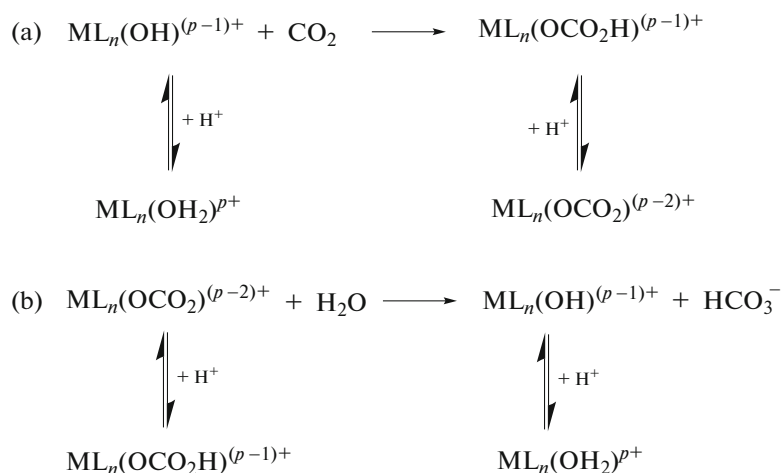
¹ The article is published in the original.

ics of the CA active site to recover some part of the enzyme's activity while imparting greater stability. Zhang and coworkers studied the kinetics of CO₂ hydrolysis catalyzed by two zinc(II) azamacrocyclic complexes: Zn-(1,4,7,10-tetrazacyclododecane) (denoted as Zn-cyclen hereafter) and Zn-(1,5,9-triazacyclododecane) (denoted as Zn-tri hereafter) [8, 9]. Both catalysts contain a zinc atom that is coordinated to nitrogen atoms (three or four) and an oxygen atom, the most prominent feature of the enzyme's catalytic site, where the nitrogen atoms belong to the histidine side chains. The intrinsic reaction rate constants of Zn-cyclen and Zn-tri were measured to be 3300 and 581 M⁻¹ s⁻¹, respectively. Although both catalysts display significantly slower CO₂ hydrolysis kinetics than HCA II, on a per mass basis they are very comparable, e.g., Zn-cyclen is less active than CA by a factor of five due to much lower molecular weight (455 Da vs 30000 Da) [10]. Koziol and coworkers studied five zinc(II) azamacrocyclic catalysts, including Zn-cyclen and Zn-tri. One of these, zinc-cyclam, had an intrinsic reaction rate constant of 5039.7 M⁻¹ s⁻¹, which is even higher than that of Zn-cyclen, the best biomimetic catalyst to the best of our knowledge. Zn-(1,4,7,10-tetraazacyclotridecane) (denoted as complex **a** hereafter), which has one more carbon atom in the ring than Zn-cyclen, also showed a quite reasonable activity with a rate coefficient of 2278.8 M⁻¹ s⁻¹. The fifth catalyst they studied was Zn-(1,4,7-triazacyclononane) (denoted as complex **b** hereafter), which has a much smaller nine-membered ring and was the least active catalyst among all five catalysts. The lowest activity was ascribed to the most weakly bound hydroxide, which may be too deactivated to effectively react with the CO₂ carbon atom.

In the 1970s, Woolley synthesized a model catalyst Zn-CR (C stands for a macrocyclic ligand, R = H or CH₃, denoted as complex **c** when R = H hereafter) to reproduce the biological properties of carbonic anhydrase [11]. This catalyst can catalyze the hydrolysis of acetaldehyde and CO₂ [11, 12]. Later, this catalyst was found to be also able to catalyze the hydrolysis of carboxylate esters and diphenyl *p*-nitrophenyl phosphate [13, 14]. In the first step of these hydrolysis reactions, the OH⁻ moiety nucleophilically attacks either the carbon or phosphorus atom of the substrate. Brown and coworkers developed a series of H₂O-bound Zn(II) complexes that resemble the configuration of HCA II's active site [15–21]. Zn-bis(1-methylimidazol-2-ylmethyl)ethylamine (denoted as complex **d** hereafter) was first introduced as an alkaline phosphatase model, and the kinetic behavior was similar to Zn-

cyclen [19, 22]. Complex **e** can catalyze CO₂ hydrolysis at pH 6.5 with a rate constant of 760 M⁻¹ s⁻¹; however, this rate constant diminishes at pH > 7 to a final value of 240 M⁻¹ s⁻¹. This unusual behavior was attributed to the formation of a less active species at higher pH [15]. Complexes **f1**, **f2**, and **f3** were introduced to overcome the relative poor binding of Zn²⁺ by complex **e** [15, 21]. ¹H NMR spectra of complex **f3** showed the appearance of a well-defined 1 : 1 complex when [Ligand **6c**]/[Zn²⁺] = 1. On the other hand, UV spectra of complexes **f1** and **f2** show little evidence of binding in the presence of Co²⁺, which was supposed to have similar behavior as Zn²⁺. As a consequence, complexes **f1** and **f2** show negligible catalysis toward CO₂ hydrolysis, while complex **f3** appears to be active. The reaction rate constant followed a distorted bell-shaped pH profile for complex **f3** maximizing at pH 6.4, which was attributed to the formation of an inactive form of the catalyst, e.g., **6c**-Zn-(OH)_x, at higher pH. Complex **g** was found to show modest catalysis of CO₂ and *p*-nitrophenylpicolinate hydrolysis at pH 6.4 and 6.7, respectively. However, the activity diminishes at pH 7.2 and disappears completely at higher values, where the insoluble ZnCO₃ was formed.

Shown in Scheme 1, carbon carbonate complexes are formed upon the reaction of metal hydroxo complexes with CO₂ and can then dissociate upon interaction with water, thereby regenerating the metal hydroxo complexes [23]. Zhang and coworkers presented a correlation between the activity of many metal hydroxo complexes and the p*K*_a value of the corresponding aqua complexes, showing that the catalytic activity of metal hydroxo complexes is enhanced linearly as p*K*_a is increased due to the increase of nucleophilicity of the bound hydroxide ion. This finding spurred us to model a series of CA mimics existing in the literature and rationalize the dependence of the activity as a function of the p*K*_a value, complementing our earlier work using DFT calculations and microkinetic modeling to study CO₂ hydrolysis by zinc-cyclen, zinc-cyclam and zinc-tri [24, 25]. The reaction rate was captured using an analytical expression that accounted for the major steps of CO₂ adsorption, HCO₃⁻ hydrolysis, and deprotonation and their relative contributions to the overall reaction rate. Utilizing this simplified PES as well as the analytical equation, screening of a variety of potential catalysts could be accelerated.



Scheme 1. CO₂ hydrolysis catalyzed by metal carbon hydroxo complexes (a) CO₂ uptake reaction, and (b) aquation reaction. M stands for metal, L stands for ligand.

Herein, we report a screening study of complexes **a** to **g** via a combination of DFT calculations and microkinetic modeling. Although no kinetic data has been reported for CO₂ hydrolysis catalyzed by complex **d**, we conjecture that this reaction can be catalyzed by complex **d**, based on comparison with hydrolysis of phosphotriesters. In addition to complexes **a** through **g**, we included zinc-cyclen, -tri, and -cyclam from previous studies in the comparison.

THEORETICAL DETAILS

Quantum chemical calculations (QC). All QC calculations were carried out using Gaussian 09 [26]. The same theoretical method that we used previously [24, 25] was applied here and is briefly summarized. Geometry optimization was carried out using the BMK functional and PCM as the solvation model, with UFF radii to create the cavity. This theoretical approach was selected based on calibration results in the literature and our own studies [27–29]. The 6-311++G(d,p) basis set was employed for all atoms except zinc, which used the LANL2DZ basis set based on previous studies [32, 33]. To locate transition state (TS) structures, the synchronous transit-guided quasi-Newton method was used. TSs were confirmed based on reaction coordinate following and eigenvalue analysis. Basis set superposition error (BSSE) was corrected according to the counterpoise method [34], using 50% of the value as recommended in the literature [35]. Free energies were calculated based on standard formulae to include zero point and temperature corrections and contributions from cavitation and dispersion–repulsion. To calculate pK_a values of each catalyst, the cycle shown in Fig. 1 was used, based on literature reports [36–40]. The formula shown in equation 1 was used, where ΔG_{gas} relies on experimental values for $G_{\text{gas}}(\text{H}^+)$ and $\Delta G_s(\text{H}^+)$ of -6.28 kcal/mol and -264.69 kcal/mol, respectively, as suggested by

Liptak and Shields [30], and ΔG_s is the solvation contribution:

$$\begin{aligned}
 pK_a &= -\log K_a = \Delta G_{\text{aq}}/2.303RT \\
 &= (\Delta G_{\text{gas}}(1\text{ M}) + \Delta G_s)/2.303RT.
 \end{aligned} \quad (1)$$

Microkinetic modeling. A microkinetic model [41] that accounted for the elementary steps studied explicitly using quantum mechanics was constructed. The rate coefficients for each reaction were calculated based on transition state theory (equation 2)

$$k(T) = \frac{k_B T}{h} (c^0)^{1-m} e^{-\frac{\Delta G^\ddagger}{RT}}. \quad (2)$$

Where k_B is Boltzmann's constant equal to 1.38×10^{-23} J K⁻¹, h is Planck's constant equal to 6.6261×10^{-34} J s, m is the number of reactants, c^0 is the standard state concentration (1 M) to which the quantum chemical calculations are referenced, and $\Delta G^\ddagger$ is the difference between the Gibbs free energy of the transition state and the reactants. The mechanism was then coupled with batch reactor design equations and solved using standard methods [42] to model CO₂ hydrolysis as a function of pH (from 5 to 12) at a temperature of 298.15 K. More specific information about the conditions for which the batch reactor model was solved is provided below.

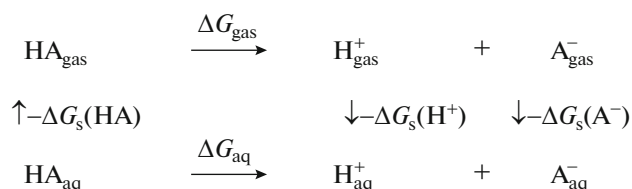


Fig. 1. Thermodynamic cycle (from Ref. 36).

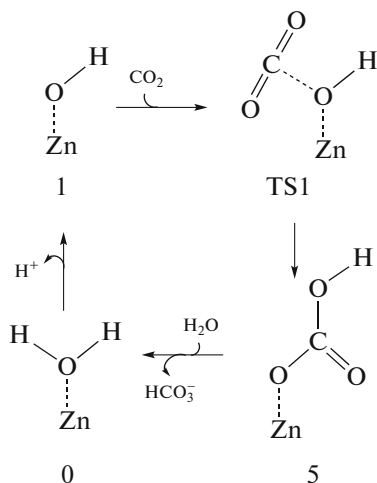
Table 1. The relative calculated free energy of structure **0**, **1**, and **TS1** for zinc-cyclen, zinc-tri, zinc-cyclam, and complex **a**, **b**, **c**, **d**, **f1**, and **f2***

	Cyclen	Tri	Cyclam	a	b	c	d	f1	f2
0	13.66	16.46	14.25	14.74	15.26	8.02	14.05	12.28	12.26
1	4.76	6.04	7.84	6.38	2.70	0.02	6.57	1.85	1.88
TS1	17.39	18.76	20.20	19.02	16.55	15.00	19.49	14.96	15.86

* Units in kcal/mol; BMK/6-311++G** (LANL2DZ basis set was used for Zn) was used for calculation; referenced to structure **5**; calculated at dead time $t = 2$ ms.

RESULTS AND DISCUSSION

Reaction mechanism. The simplified catalytic reaction cycle is mapped in Scheme 2. Species are numbered according to reference 24 for an easier comparison with our previous studies. The structures of ligand **a** to **g** are shown in Fig. 2. The free energy profile for complex **a** is shown in Fig. 3 as an illustrative example. The relative free energies for the different catalysts are tallied in Table 1. The more detailed reaction mechanism was described previously and is very much like that of CA [24, 25]. In the first step of the mechanism, CO_2 nucleophilically attacks via its carbon atom the oxygen atom of zinc-hydroxide (structure **1**), and a weakly associated precursor complex is formed via a large barrier (**TS1**). The barrier, which determines how fast CO_2 is adsorbed to the reaction center, is regarded as one of the important features along the potential energy surface (PES). Koziol and coworkers studied five zinc(II) azamacrocyclic complexes and suggested that the more electron-donating ligands were the larger ones and therefore led to weaker Lewis acidity of Zn^{2+} and a “freer” OH^- [31]. A “free” hydroxide will undoubtedly facilitate CO_2 adsorption. The extreme case would be OH^- itself, which is the freest hydroxide. Even the activity measures of the best biomimetic catalysts, zinc-cyclen ($3300 \text{ M}^{-1} \text{ s}^{-1}$) and zinc-cyclam ($5039 \text{ M}^{-1} \text{ s}^{-1}$), are still lower than that of $\text{OH}^-(\text{aq.})$ (7900 M^{-1}) [8, 31].

**Scheme 2.** Simplified reaction mechanism for CO_2 hydrolysis catalyzed by CA mimics. Species are numbered according to [24].

The weakly associated precursor complex will then form a bicarbonate bound species, which will subsequently rearrange to a better leaving group (structure **5**) via a series of intermediates and small barriers. Structure **5** was shown previously to be the one with the lowest free energy along the reaction coordinate for zinc-cyclen, zinc-cyclam, zinc-tri, and cobalt-cyclen, also consistent with investigations of other carbonic anhydrase mimics [25, 32–34]. After the formation of a better leaving group, the bound bicarbonate anion is displaced by H_2O from the surrounding medium, and the resulting product is a zinc-water species (structure **0**) and a free bicarbonate anion. The barrier for this step was found to be highly related to steric effects due to a change in the coordination number [25].

The final step is the deprotonation of the zinc-water species (structure **0**) and the regeneration of the zinc-hydroxide species (structure **1**). This step is widely regarded as the rate-limiting step for CA-catalyzed CO_2 hydrolysis [35–37], while the first step, CO_2 adsorption, is the rate-determining step for biomimetic catalysts [8, 9, 24, 25]. In a CA-catalyzed system, an effective pre-association of CO_2 with the hydrophobic pocket (binding constant 10^4 M^{-1}) will be formed and assist the CO_2 adsorption step [6, 9]. Thus, even the activity of the best biomimetic catalysts, zinc-cyclen ($3300 \text{ M}^{-1} \text{ s}^{-1}$) and zinc-cyclam ($5039 \text{ M}^{-1} \text{ s}^{-1}$), is still orders of magnitude below that of HCA II.

pK_a values. The calculated and experimental pK_a values of various carbonic anhydrase mimics are summarized in Table 2. Note that the experimental pK_a values of complexes **b**, **f1**, **f2**, and **f3** have not, to the best of our knowledge, been reported. We previously demonstrated very good agreement between experimental and calculated pK_a values for zinc-ligand complexes, and good agreement was also obtained for the new complexes examined here. Complex **a**, which has one more carbon in the ring than zinc-cyclen, was calculated to have a pK_a value of 8.3, while the experimental pK_a value was also determined to be 8.3 [31]. Considering zinc-cyclam (expt. pK_a 9.8 versus calc. pK_a 9.7) and zinc-tri (expt. pK_a 7.3 versus calc. pK_a 6.8), the method applied here is surprisingly accurate in predicting pK_a values of zinc complexes with azamacrocyclic ligands. Ligand **b** also has an aza-macro-

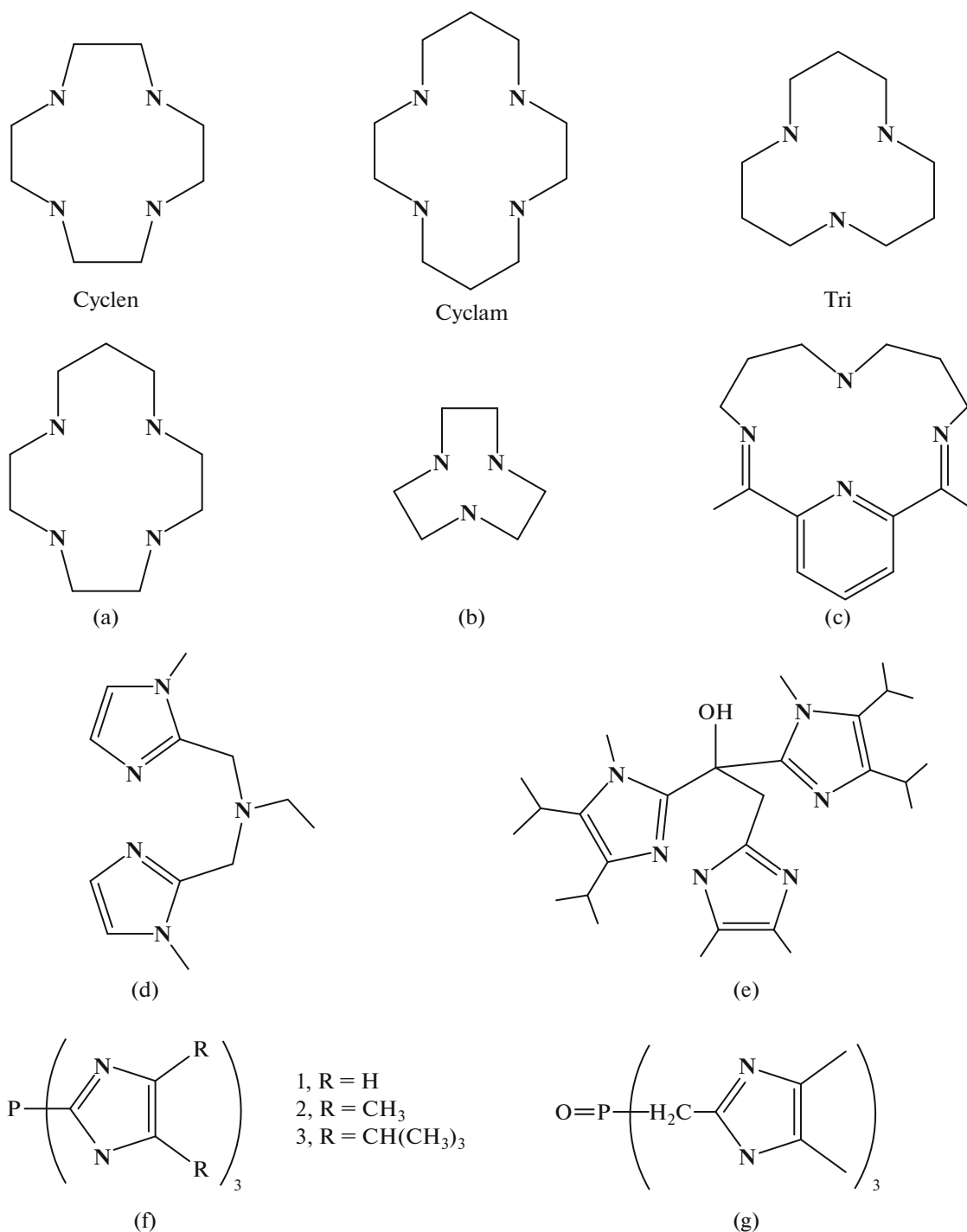


Fig. 2. Structures of various ligands.

cyclic structure. The pK_a value of complex **b** was calculated to be 5.2, which is in good agreement with previous calculated values of 5.6 and 4.3 [31]. Ligand **c** is the only aromatic aza-macrocyclic ligand studied in the current work. The pK_a value of complex **c** was calculated to be 9.6, slightly higher than the experimental value of 8.7 [11].

Complexes **d**, **e**, **f**, and **g** all contain imidazoles, which is very similar to the wild type enzyme. The active site of HCA II is comprised of a Zn^{2+} ion coordinated to three histidine imidazoles and an OH^- anion. Unlike complexes **e**, **f**, and **g**, which all have three imidazoles, complex **d** has only two imidazoles. The pK_a value of complex **d** was calculated as 9.0,

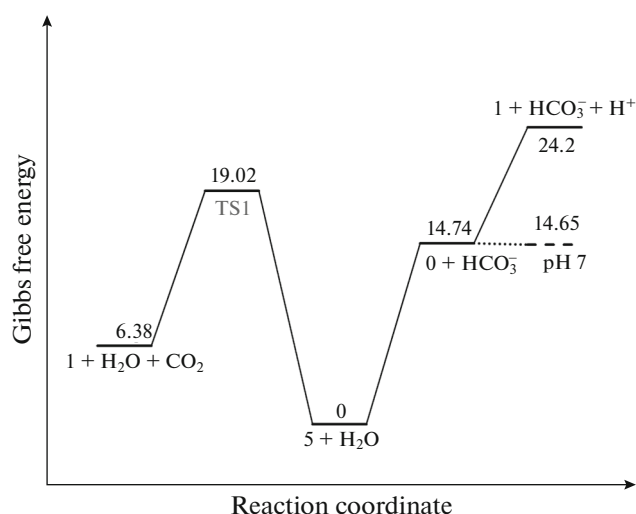
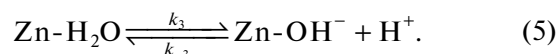
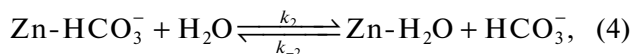
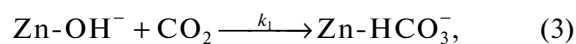


Fig. 3. Simplified version of the free energy surface for complex **a**. Species are numbered according to the labels used in Scheme 2. Units are in kcal/mol. (The dashed line denotes the free energy of the final state if a buffer of pH 7 were used).

while the experimental value was 9.2 at 0°C [12]. The experimental value should be smaller at room temperature, since the deprotonation process is endothermic. For example, quantitative titration studies showed that the pK_a value of a 1:1 mixture of ligand **d** and Zn^{2+} was about 8.2 at room temperature [19]. The pK_a value of complex **e** was calculated to be 7.0, while no typical pH profile was observed experimentally due to poor binding of Zn^{2+} . Ligand **e** in the presence of an equivalent amount of Zn^{2+} shows reversible consumption of equivalent OH^- with an apparent pK_a of 6.5 in 76% ethanolic H_2O [15]. However, the pK_a value reduced to 5.8 in a less polar medium, e.g. 60% aqueous dioxane. For complexes **f1**, **f2**, and **f3**, no experimental pK_a values have been reported, while the calculated values are 6.8, 6.8, and 5.6, respectively. The pK_a value of complex **g** was

calculated to be 5.5, while titration showed that 1:1 ligand **g**: Zn^{2+} has a pK_a value of 6.0.

Intrinsic Reaction Rate Constant. In simplified form, the reaction mechanism is summarized below as three main steps:



Using equation 2, rate coefficients were calculated for each of the three steps, and then microkinetic modeling was carried out, providing the overall reaction rate as a function of pH. Since only the $Zn-OH^-$ form catalyzes the hydrolysis reaction, the intrinsic reaction rate constant was calculated via the equation:

$$k_{int} = k_{cat}(1 + 10^{pK_a - pH}). \quad (6)$$

The approach used experimentally to measure the reaction rate coefficient was described previously [24, 25], and the inclusion of a reversible buffer reaction:



in the microkinetic model to mimic intramolecular and intermolecular proton transfer [50, 51] was also described previously [24, 25].

The calculated and experimental intrinsic reaction rate constants for all catalysts are listed in Table 2. For zinc-cyclen, zinc-tri, and zinc-cyclam, we have previously performed microkinetic modeling on the basis of the complete reaction mechanism. It would be very valuable to compare these results with current microkinetic modeling results on the basis of the simplified reaction mechanism, and test the reliability of current methods. Previously, we reported that the intrinsic reaction rate constant for zinc-cyclen was $3055 \text{ M}^{-1} \text{ s}^{-1}$, in very good agreement with the experimental values of 3300 and $2691.5 \text{ M}^{-1} \text{ s}^{-1}$ [8, 31]. With the simplified reaction mechanism the intrinsic reaction rate constant of zinc-cyclen was calculated to be

Table 2. The calculated and experimental pK_a values and intrinsic reaction rate constants for zinc-tri, zinc-cyclen, zinc-cyclam, and complexes **a** through **g** at 298 K

	Cyclen	Tri	Cyclam	a	b	c	d	e	f1	f2	f3	g
calc. pK_a	7.9	6.8	9.7	8.3	5.2	9.6	9.0	7.0	6.8	6.8	5.6	5.5
expt. pK_a	8.1	7.3	9.8	8.3	—	8.7	8.2	6.5	—	—	—	6.0
calc. k_{int}^*	3051	2695	4583	3089	425	60	1987	—	1340	339	—	—
expt. k_{int}^*	2692	1104	5040	2279	88	25	—	760	—	—	900	156

* Units in $\text{M}^{-1} \text{ s}^{-1}$; BMK/6-311++G** (LANL2DZ basis set was used for Zn) was used for calculation.

$3051 \text{ M}^{-1} \text{ s}^{-1}$, only slightly smaller than that from the complete reaction mechanism. The intrinsic reaction rate constant for zinc-tri was previously calculated to be $2693 \text{ M}^{-1} \text{ s}^{-1}$, which compares relatively well to, albeit slightly higher than, the experimental values of $581 \text{ M}^{-1} \text{ s}^{-1}$ reported by Zhang et al. and $1104.2 \text{ M}^{-1} \text{ s}^{-1}$ reported by Koziol et al. [9, 31]. Again, with the simplified reaction mechanism, the intrinsic reaction rate constant of zinc-tri was calculated to $2695 \text{ M}^{-1} \text{ s}^{-1}$, only slightly higher than that from the complete reaction mechanism. For zinc-cyclam, an intrinsic reaction rate coefficient of $4623 \text{ M}^{-1} \text{ s}^{-1}$ was obtained, in very good agreement with that predicted by Koziol et al. of $5039.7 \text{ M}^{-1} \text{ s}^{-1}$ [31]. Similarly, with the simplified reaction mechanism, the intrinsic reaction rate constant of zinc-cyclam was calculated to $4583 \text{ M}^{-1} \text{ s}^{-1}$, slightly smaller than that from the complete reaction mechanism. Based on these comparisons, the simplified mechanism is able to recapitulate the intrinsic reaction rate constants well.

Koziol and coworkers studied reaction kinetics of five zinc complexes of aza-macrocyclic ligands, including zinc-cyclen, zinc-tri, and zinc-cyclam. The other two catalysts they studied were complexes **a** and **b**. Utilizing microkinetic modeling based on the simplified reaction mechanism, the intrinsic reaction rate constant of complex **a** was calculated to be $3089 \text{ M}^{-1} \text{ s}^{-1}$, in good agreement with the experimental value of $2278.8 \text{ M}^{-1} \text{ s}^{-1}$. Comparing to zinc-cyclen, complex **a** has one more carbon atom and a slightly higher $\text{p}K_{\text{a}}$ value (only 0.2 units higher), and thus complex **a** is expected to show very similar behavior to zinc-cyclen. As shown in Fig. 4, the sigmoidal shape of complex **a** is indeed very much like that of zinc-cyclen, with the inflection point moved to pH 8.3, which is identical to the calculated $\text{p}K_{\text{a}}$ value of complex **a**. For complex **b**, the intrinsic reaction rate constant was computed to be $425 \text{ M}^{-1} \text{ s}^{-1}$, slightly higher than the experimental value of $88.3 \text{ M}^{-1} \text{ s}^{-1}$. The intrinsic reaction rate constants of zinc-cyclen, zinc-cyclam, and complex **a** show excellent agreement with the experimental values, while the intrinsic reaction rate constants of zinc-tri and complex **b** are higher than the experimental values.

The discrepancies between the experimental and calculated reaction rate constants for zinc-tri and complex **b** need to be investigated. As indicated by Kimura and others, zinc-tri and complex **b** are not stable at pH values above 8 and tend to precipitate [38]. Zhang and van Eldik summarized the stability constants of various carbonic anhydrase mimics, and attributed the high activity of zinc-cyclen to the significantly high stability constant [8]. To compare the relative stability of these five aza-macrocyclic com-

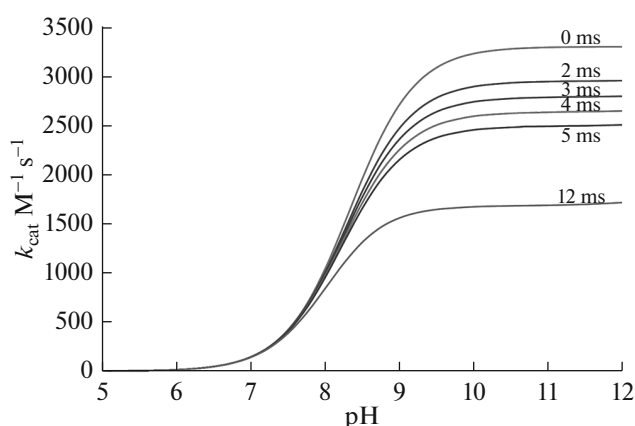


Fig. 4. Reaction rate constant of CO_2 hydrolysis by complex **a** versus pH based on the rate from the microkinetic model at $t = 0, 2, 3, 4, 5$, and 12 ms .

plexes, we calculated binding energies of OH^- and Zn^{2+} , which are listed in Table 3:

$$\Delta G_{\text{binding}}(\text{OH}^-) = G([\text{Ligand-Zn-OH}]^+) - G([\text{Ligand-Zn}]^{2+}) - G(\text{OH}^-), \quad (8)$$

$$\Delta G_{\text{binding}}(\text{Zn}^{2+}) = G([\text{Ligand-Zn}]^{2+}) - G(\text{Ligand}) - G(\text{Zn}^{2+}). \quad (9)$$

Zinc-cyclam has the highest binding energy of OH^- , while complex **b** has the lowest, indicating OH^- anion binds complex **b** the most tightly. Interestingly, the trends of binding energy of OH^- may be related to the $\text{p}K_{\text{a}}$ values. As shown in Fig. 5, $\Delta G_{\text{binding}}(\text{OH}^-)$ increases linearly with the $\text{p}K_{\text{a}}$ value, and a linear equation can be obtained by fitting these data:

$$\Delta G_{\text{binding}}(\text{OH}^-) = 1.661\text{p}K_{\text{a}} - 41.623. \quad (10)$$

The binding energies of OH^- for zinc-cyclen and zinc-tri were calculated to be -27.59 and -29.59 kcal/mol , respectively, while the stability constant for zinc-cyclen (15.7) is significantly larger than that for zinc-tri (1.1). Thus, the relative stability cannot be correlated with the binding energy of OH^- . By contrast, the binding energies of Zn^{2+} of zinc-cyclam, zinc-cyclen, and complex **a** are very close, and much smaller than those of zinc-tri and complex **b**, suggesting the cyclam, cyclen, and **a** ligands would bind Zn^{2+} much more tightly than the tri and **b** ligands. The low stability constants for zinc-tri and complex **b** thus appear to be correlated with the poor binding of Zn^{2+} , and therefore the experimental activity is lower than that predicted by our model.

What do these trends mean in terms of the coordination number around Zn^{2+} ? The higher stabilities of zinc-cyclen, zinc-cyclam, and complex **a** may be related to the fact that they are pentacoordinated.

Table 3. The calculated binding free energies of OH[−] and Zn²⁺ for zinc-cyclen, zinc-tri, zinc-cyclam and complexes **a**, **b**, **f1**, and **f2** at 298 K^a

	Cyclen	Tri	Cyclam	a	b	f1	f2
$\Delta G_{\text{binding}}(\text{OH}^-)^{\text{b}}$	−27.59	−29.59	−25.69	−28.48	−33.31	−31.36	−30.22
$\Delta G_{\text{binding}}(\text{Zn}^{2+})^{\text{c}}$	−72.00	−57.36	−76.05	−68.87	−53.97	−48.63	−54.26

^a Units in kcal/mol; BMK/6-311++G** (LANL2DZ basis set was used for Zn) was used for calculation. ^b $\Delta G_{\text{binding}}(\text{OH}^-) = G([\text{Ligand-Zn-OH}]^+) - G([\text{Ligand-Zn}]^{2+}) - G(\text{OH}^-)$. ^c $\Delta G_{\text{binding}}(\text{Zn}^{2+}) = G([\text{Ligand-Zn}]^{2+}) - G(\text{Ligand}) - G(\text{Zn}^{2+})$.

Interestingly, like zinc-tri and complex **b**, HCA II is also tetracoordinated. Experiments showed that the Zn^{II} ion can be easily extracted from HCA II without denaturation of the enzyme [39]. Nevertheless, the tetracoordinated structure favors bicarbonate displacement by water, in which a pentacoordinated transition state will be formed. We have previously shown that compared to zinc-cyclen and zinc-cyclam, zinc-tri has the smallest barrier for this step, perhaps due to a more stable transition state [25]. In the case of zinc-tri and complex **b**, the relative instability may also be related to ring strain. For zinc-tri and complex **b**, the zinc atom is markedly higher than the ring plane, while for zinc-cyclen, zinc-cyclam, and complex **a**, the zinc atom resides in the ring plane.

The intrinsic reaction rate constant of complex **c** was calculated to be 60 M^{−1} s^{−1} at 25°C. This is in good agreement with available experimental information, with an experimental rate coefficient of 25 M^{−1} s^{−1} at 0°C, and a higher value would be expected if the temperature was 25°C [11]. For complex **d**, although no experimental data was reported for the hydrolysis of CO₂, the intrinsic reaction rate constant was predicted to be 1987 M^{−1} s^{−1}. Complexes **f1**, **f2**, and **f3** have very similar structures. In experiment, complexes **f1** and **f2** showed negligible activity toward CO₂ hydrolysis due

to ineffective complexation with Zn²⁺, while complex **f3** shows a well-defined 1 : 1 complex when [ligand **f3**] : [Zn²⁺] = 1 : 1, with an intrinsic reaction rate constant of about 472–900 M^{−1} s^{−1} [15]. While the measured activities of complexes **f1** and **f2** are negligible due to poor complexation, it is still worthwhile to model them and enrich our inventory of catalysts. The intrinsic reaction rate constant of complex **f1** was computed to be 1340 M^{−1} s^{−1}, while the intrinsic reaction rate constant of complex **f2** was calculated to be 339 M^{−1} s^{−1}. Both of the predicted intrinsic reaction rate constants are in reasonable agreement with complex **f3**'s activity. The activity of complex **f1** is about four times higher than that of complex **f2**, due to reduced steric hindrance for CO₂ adsorption by complex **f1**. The binding energies of OH[−] and Zn²⁺ for complexes **f1** and **f2** are listed in Table 1. $\Delta G_{\text{binding}}(\text{Zn}^{2+})$ for complexes **f1** and **f2** are within the range of those for zinc-tri and complex **b** and significantly higher than those of zinc-cyclen, zinc-cyclam, and complex **a**. Therefore, it is not surprising that complexes **f1** and **f2** are not stable. On the other hand, the binding energies of OH[−] for these complexes fit in equation 10 very well, indicating that this equation may be able to expand to non-azamacrocyclic ligands.

Reaction rate coefficient decreases with increasing dead time. It is important to reconcile the calculated rate coefficients with the fact that the measured reaction rate coefficient is a function of the “dead time” of the stopped-flow spectroscopic measurement technique, as described in our previous work [24]. We have previously shown that the reaction rate constant decreased to 50% of its initial value for zinc-cyclen and zinc-tri and 33% of its initial value for zinc-cyclam when the time was increased from 0 to 12 ms. Similar behavior was observed when the simplified reaction mechanism was used for these three catalysts, suggesting that the simplified mechanism is sufficient to capture the essential behavior of the complete reaction mechanism. The pK_a values and intrinsic reaction rate constants of complex **a** and zinc-cyclen are very close, and their structures are also very similar. Therefore, it is not surprising to see both catalysts behave very similarly between 0 to 12 ms, as shown in Fig. 4. Complex **b** is the least active among the five aza-macrocyclic complexes. However, as shown in Fig. 6, from

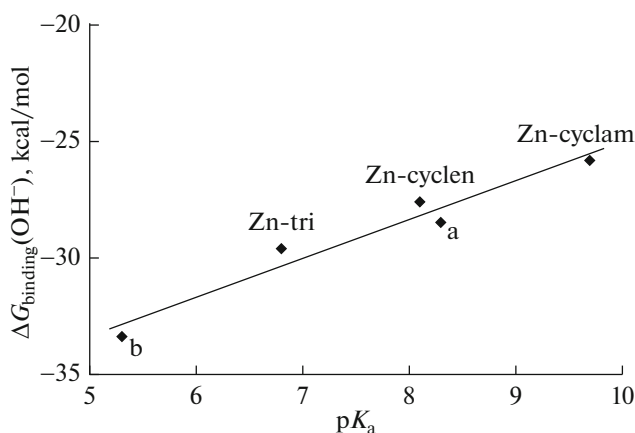


Fig. 5. The binding free energy of OH[−] ($\Delta G_{\text{binding}}(\text{OH}^-) = G([\text{Ligand-Zn-OH}]^+) - G([\text{Ligand-Zn}]^{2+}) - G(\text{OH}^-)$) plotted as a function of the pK_a value.

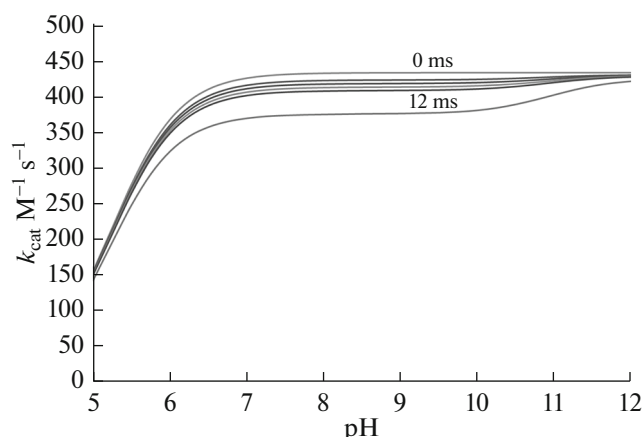


Fig. 6. Reaction rate constant of CO₂ hydrolysis by complex **b** versus pH based on the rate from the microkinetic model at $t = 0$ (upper most curve), 2, 3, 4, 5, and 12 (lowest curve) ms.

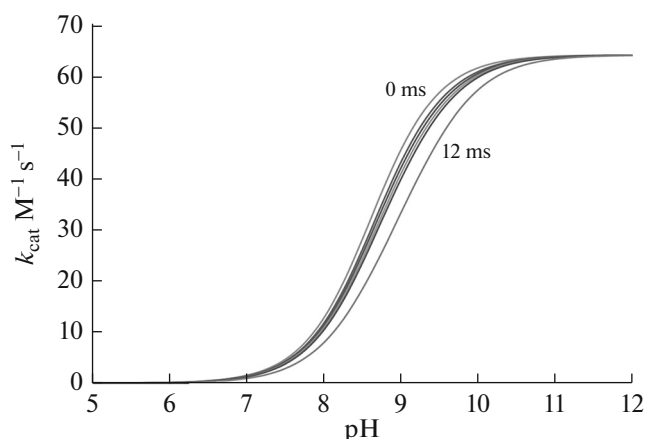
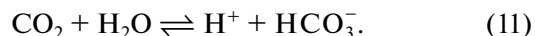


Fig. 7. Reaction rate constant of CO₂ hydrolysis by complex **c** versus pH based on the rate from the microkinetic model at $t = 0$ (upper most curve), 2, 3, 4, 5, and 12 (lowest curve) ms.

0 to 12 ms its reaction rate constant is nearly unchanged, due to a very small energy difference between **1** and **5**, 2.70 kcal/mol, the smallest among all five aza-macrocyclic complexes. By tracking the concentration profiles of intermediates, we identified that only a very small portion of the catalytic form, **1**, is converted to **5**. Complex **c** also has a macrocyclic structure and has almost same free energy between **1** and **5**, with only a 0.02 kcal/mol difference. The reaction rate constant curve at 0 ms is almost identical to that at 12 ms, as shown in Fig. 7. The tiny difference is due to the consumption of carbon dioxide and conversion of **1** to **5**. Although no kinetic data of CO₂ hydrolysis was reported for complex **d**, its activity was predicted to be quite high. As shown in Fig. 8, from 0 to 12 ms the reaction rate constant of complex **d** decreased to two-thirds of its initial value, due to the conversion of **1** to **5**.

For complexes **f1** and **f2**, the energy difference between **1** and **5** is very close, 1.85 and 1.88 kcal/mol, respectively. As shown in Fig. 9, the reaction rate constant of complex **f2** decreased only slightly as expected as time increased from 0 to 12 ms; however, the reaction rate constant of complex **f1** decreased to one-third of its initial value for pH 7 to 10 as shown in Fig. 10, and the curve of reaction rate constant shows an unusual double sigmoidal shape. By tracking the concentration profiles of all species in our microkinetic model, we found that the concentration of **1** decreased sharply over the middle range of pH, which was converted to **5**. As discussed previously, the net reaction for all three catalysts is the same and can be expressed as:



Therefore, the intrinsic reaction free energy for carbon dioxide hydrolysis is also the same regardless of the catalyst. As illustrated in Fig. 3, by maintaining the

concentration of H⁺ or pH, the practical effect is to lower the height of **1** at the end of free energy landscape. The amount that it is lowered is readily calculated by $RT\ln(1/[\text{H}^+])$ or 1.364 pH kcal/mol at 298.15 K. For example, a buffer with pH 7 would reduce the height of the final step by 9.55 kcal/mol. For complex **f1**, the energy difference between **5** and **1** + HCO₃⁻ + H⁺ is also very small in the high pH region, and thus compared to the middle pH region, less **1** was converted to **5**. Note that complex **f2** has a very similar free energy profile but a much smaller reaction constant, and therefore the same double sigmoidal curve would have been observed if the reaction time were longer.

Variation of the intrinsic reaction rate constant with pK_a . The activity of carbonic anhydrase mimics was found to be highly related to pK_a [8, 12, 23]. For example, in a study of CO₂ hydrolysis by metal hydroxo

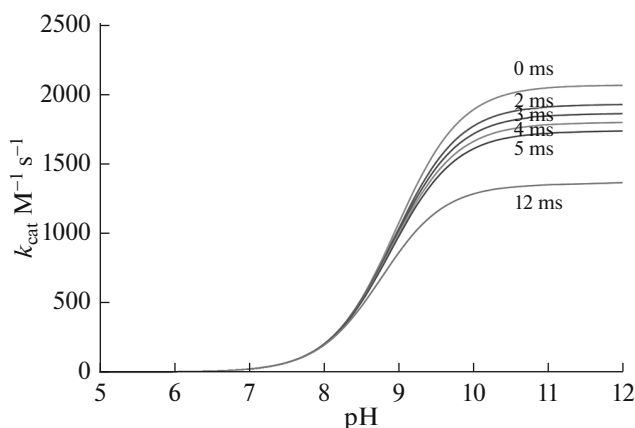


Fig. 8. Reaction rate constant of CO₂ hydrolysis by complex **d** versus pH based on the rate from the microkinetic model at $t = 0$, 2, 3, 4, 5, and 12 ms.

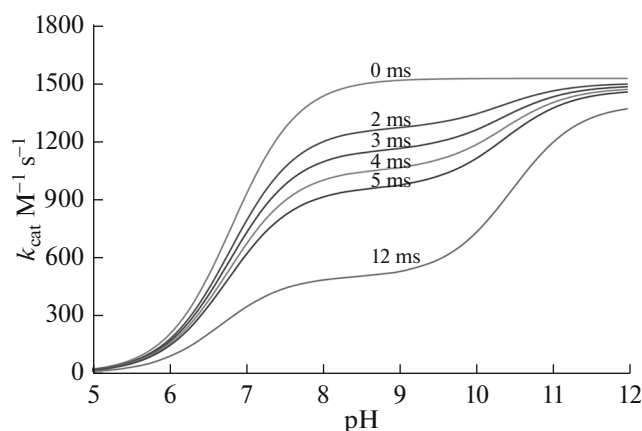


Fig. 9. Reaction rate constant of CO_2 hydrolysis by complex **f1** versus pH based on the rate from the microkinetic model at $t = 0, 2, 3, 4, 5$, and 12 ms.

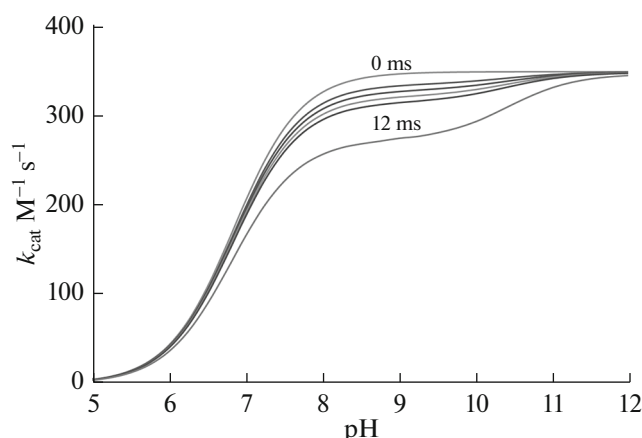


Fig. 10. Reaction rate constant of CO_2 hydrolysis by complex **f2** versus pH based on the rate from the microkinetic model at $t = 0$ (upper most curve), $2, 3, 4, 5$, and 12 (lowest curve) ms.

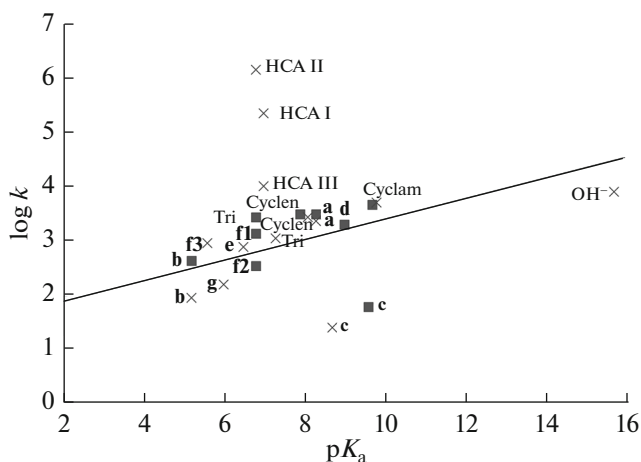


Fig. 11. Plot of $\log k$ from experimental (X) and calculated (square) results versus $\text{p}K_a$ for the hydrolysis of CO_2 by various catalysts. The experimental data for HCA I, HCA II, and HCA III were obtained from Ref. 8.

complexes, the rate constant was found to increase linearly with the $\text{p}K_a$ of $\text{ML}_n(\text{OH})_2^{p+}$, which is ascribed to the increase of nucleophilicity of the bound hydroxide ion [23]. Other carbonic anhydrase mimics were also found to obey this correlation [8, 12]. We have shown that the $\text{p}K_a$ value was largely correlated to the binding free energy of OH^- . The higher the $\text{p}K_a$ value, the larger the binding free energy, and therefore the “freer” the hydroxide. An extreme example is OH^- (aq.), the freest hydroxide, whose activity is higher than any known biomimetic catalyst [8, 31]. We present a correlation between the activity of carbonic anhydrase mimics studied in this work with $\text{p}K_a$, as shown in Fig. 11. The trend line was taken from an earlier correlation [23]. With the exception of complex **c**, the experimental and calculated reaction rate constants fit the correlation very well. The unusually low rate constant of complex **c** is probably due to the expulsion between the p-electrons of the oxygen atom in CO_2 and the π -electrons of pyridine, which will increase the energy of **TS1**.

It is very interesting to note that in a study of various mutants of HCA II a decrease of activity was found with an increase in the $\text{p}K_a$ value [40]. Unlike carbonic anhydrase mimics, the rate-limiting step for CO_2 hydrolysis by HCA II is the deprotonation of the $\text{Zn-H}_2\text{O}$ species. The lower the $\text{p}K_a$ value, the more unstable the $\text{Zn-H}_2\text{O}$ species, and thus more favorable the deprotonation step. On the other hand, the proportional relationship between the activity and $\text{p}K_a$ value of carbonic anhydrase confirmed our previous conclusion that the deprotonation step is not rate-limiting, and also indicated the adsorption of CO_2 is very likely to be the rate-limiting step due to the increase of nucleophilicity.

CONCLUSIONS

Six carbonic anhydrase mimics were studied using density functional theory. Thermodynamic and kinetic parameters were calculated for a catalytic cycle comprised of three major steps in the reaction mechanism. The results were directly compared with our previous results for zinc-cyclen, zinc-cyclam, and zinc-tri. The $\text{p}K_a$ values of these six mimics and three additional mimics were calculated and were in good agreement with experimental results. The $\text{p}K_a$ values of five zinc(II) aza-macrocyclic complexes were found to be correlated to their binding free energy of OH^- , and an equation was obtained to correlate the data, which was subsequently found to be applicable to complexes **f1** and **f2**. Using kinetic and thermodynamic parameters derived from first principles, microkinetic modeling was conducted. Among five zinc(II) aza-macrocyclic complexes, the intrinsic rate constants of zinc-tri and complex **b** were overestimated, which might be due to their instability at high

pH. The reaction rate constant also depends on the availability of the catalyst. Over 0 to 12 ms, the observed reaction rate constant for some mimics decreased, and this was attributed to the decrease of the concentration of the catalytic form, Zn-OH^- , which was primarily converted to Zn-HCO_3^- . From 0 to 12 ms, the rate constant of complexes **a** and **d** decreased to 50% and 67% of their initial values, respectively; the rate constants of complexes **b** and **f2** were almost unchanged, while the rate constant of complex **f1** showed an unusual double sigmoidal shape. Finally, we present a correlation between the activity of many carbonic anhydrase mimics and their K_a values. The proportional relationship indicates that the adsorption of CO_2 is the rate-limiting step, consistent with full analyses of the rate-limiting step via microkinetic modeling that we did earlier for a subset of these catalysts.

ACKNOWLEDGMENTS

The authors are grateful for the financial support provided by ConocoPhillips and Phillips 66.

NOTATION

$\Delta G_{\text{binding}}$	free energy of binding
ΔG_{aq}	free energy change in aqueous solution
ΔG_{gas}	free energy in the gas phase
k_{int}	intrinsic rate coefficient
k_{cat}	rate coefficient of overall catalytic cycle
k_B	Boltzmann's constant ($1.38 \times 10^{23} \text{ J K}^{-1}$)
h	Planck's constant ($6.6261 \times 10^{-34} \text{ J s}$)
m	number of reactants
c^0	standard state concentration (1 M)
$\Delta G^\ddagger$	difference between the Gibbs free energy of the transition state and the reactants

REFERENCES

- Montzka, S.A., Dlugokencky, E.J., and Butler, J.H., Non- CO_2 greenhouse gases and climate change, *Nature*, 2011, vol. 476, no. 7358, pp. 43–50. doi 10.1038/nature10322
- Maiti, A., Atomistic modeling toward high-efficiency carbon capture: A brief survey with a few illustrative examples, *Int. J. Quantum Chem.*, 2014, vol. 114, no. 3, pp. 163–175. doi 10.1002/qua.24579
- Petit, J.R., Jouzel, J., Raynaud, D., Barkov, N.I., Barnola, J.M., Basile, I., Bender, M., Chappellaz, J., Davis, M., Delaygue, G., Delmotte, M., Kotlyakov, V.M., Legrand, M., Lipenkov, V.Y., Lorius, C., et al., Climate and atmospheric history of the past 420,000 years from the Vostok ice core, Antarctica, *Nature*, 1999, vol. 399, no. 6735, pp. 429–436. doi 10.1038/20859
- Figuerola, J.D., Fout, T., Plasynski, S., McIlvried, H., and Srivastava, R.D., Advances in CO_2 capture technology – The U.S. Department of Energy's Carbon Sequestration Program, *Int. J. Greenh. Gas Control*, 2008, vol. 2, no. 1, p. 9.
- Bao, L.H. and Trachtenberg, M.C., Facilitated transport of CO_2 across a liquid membrane: Comparing enzyme, amine, and alkaline, *J. Membr. Sci.*, 2006, vol. 280, nos. 1–2, p. 330.
- Sjoblom, B., Polentarutti, M., and Djinicic-Carugo, K., Structural study of X-ray induced activation of carbonic anhydrase, *Proc. Natl. Acad. Sci. U. S. A.*, 2009, vol. 106, no. 26, p. 10609.
- Domsic, J.F., Avvaru, B.S., Kim, C.U., Gruner, S.M., Agbandje-McKenna, M., Silverman, D.N., and McKenna, R., Entrapment of carbon dioxide in the active site of carbonic anhydrase II, *J. Biol. Chem.*, 2008, vol. 283, no. 45, p. 30766.
- Zhang, X. and van Eldik, R., A functional model for carbonic anhydrase: Thermodynamic and kinetic study of a tetraazacyclododecane complex of zinc(II), *Inorg. Chem.*, 1995, vol. 34, no. 22, pp. 5606–5614. doi 10.1021/ic00126a034
- Zhang, X., van Eldik, R., Koike, T., and Kimura, E., Kinetics and mechanism of the hydration of carbon dioxide and dehydration of bicarbonate catalyzed by a zinc (II) complex of 1,5,9-triazacyclododecane as a model for carbonic anhydrase, *Inorg. Chem.*, 1993, vol. 32, no. 25, pp. 5749–5755. doi 10.1021/ic00077a017
- Floyd, W.C., Baker, S.E., Valdez, C.A., Stolaroff, J.K., Bearinger, J.P., Satcher, J.H., and Aines, R.D., Evaluation of a carbonic anhydrase mimic for industrial carbon capture, *Environ. Sci. Technol.*, 2013, vol. 47, no. 17, p. 10049.
- Woolley, P., Models for metal-ion function in carbonic-anhydrase, *Nature*, 1975, vol. 258, no. 5537, p. 677.
- Woolley, P., Metal-ion assisted catalysis of nucleophilic attack: 3. Cooperation between metal-ion and hydroxide, *J. Chem. Soc., Perkin Trans. 2*, 1977, vol. 2, no. 3, p. 318.
- Gellman, S.H., Petter, R., and Breslow, R., Catalytic hydrolysis of a phosphate triester by tetracoordinated zinc complexes, *J. Am. Chem. Soc.*, 1986, vol. 108, no. 9, pp. 2388–2394. doi 10.1021/ja00269a041
- Chin, J. and Zou, X., Relationship between effective nucleophilic catalysis in the hydrolysis of esters with poor leaving groups and the lifetime of the tetrahedral intermediate, *J. Am. Chem. Soc.*, 1984, vol. 106, no. 12, pp. 3687–3688. doi 10.1021/ja00324a056
- Huguet, J. and Brown, R.S., Catalytically active models for the active site in carbonic anhydrase, *J. Am. Chem. Soc.*, 1980, vol. 102, no. 25, pp. 7571–7572. doi 10.1021/ja00545a034
- Brown, R.S., Salmon, D., Curtis, N.J., and Kusuma, S., Carbonic anhydrase models. 4. [Tris[(4,5-dimethyl-2-imidazolyl)methyl]phosphine oxide]cobalt(2+), a small-molecule mimic of the spectroscopic properties of cobalt(II) carbonic anhydrase, *J. Am. Chem. Soc.*, 1982, vol. 104, no. 11, pp. 3188–3194. doi 10.1021/ja00375a043

17. Brown, R.S., Trisimidazolylphosphine: M(II); Models for the metal-binding site in carbonic anhydrase, *Inorg. Chim. Acta*, 1983, vol. 79, pp. 64–65. doi 10.1016/S0020-1693(00)95091-6
18. Slebocka-Tilk, H., Cocho, J.L., Frackman, Z., and Brown, R.S., Carbonic anhydrase models. 5. Tris(4,5-di-normal-propyl-2-imidazolyl)phosphine-zinc(2+) and Bis(4,5-diisopropyl-2-imidazolyl)-2-imidazolylphosphine-zinc(2+). Catalysts Facilitating HCO_3^- Reversible CO_2 Interconversion, *J. Am. Chem. Soc.*, 1984, vol. 106, no. 8, pp. 2421–2431. doi 10.1021/ja00320a032
19. Clewley, R.G., Slebockatilk, H., and Brown, R.S., Mononuclear and dinuclear M-2+ chelates as catalysts for the hydrolysis of organo-phosphate triesters, *Inorg. Chim. Acta*, 1989, vol. 157, no. 2, p. 233.
20. Brown, R.S. and Huguet, J., Synthesis and physical studies of pyridine and imidazole containing tridentate metal binding ligands, *Can. J. Chem.*, 1980, vol. 58, no. 9, pp. 889–901. doi 10.1139/v80-141
21. Brown, R.S., Curtis, N.J., and Huguet, J., Tris(4,5-diisopropylimidazol-2-yl)phosphine:zinc(2+). A catalytically active model for carbonic anhydrase, *J. Am. Chem. Soc.*, 1981, vol. 103, no. 23, pp. 6953–6959. doi 10.1021/ja00413a031
22. Koike, T. and Kimura, E., Roles of zinc(II) ion in phosphatases. A model study with zinc(II)-macrocyclic polyamine complexes, *J. Am. Chem. Soc.*, 1991, vol. 113, no. 23, pp. 8935–8941. doi 10.1021/ja00023a048
23. Palmer, D.A. and van Eldik, R., The chemistry of metal carbonate and carbon dioxide complexes, *Chem. Rev.*, 1983, vol. 83, no. 6, pp. 651–731. doi 10.1021/cr00058a004
24. Ma, R., Scheutte, G.F., and Broadbelt, L.J., Microkinetic modeling of CO_2 hydrolysis over Zn-(1,4,7,10-tetraazacyclododecane) catalyst based on first principles: Revelation of rate-determining step, *J. Catal.*, 2014, vol. 317, p. 176.
25. Ma, R., Scheutte, G.F., and Broadbelt, L.J., Insights into the relationship of catalytic activity and structure: a comparison study of three carbonic anhydrase mimics, *Int. J. Chem. Kinet.*, 2014, vol. 46, no. 11, p. 683.
26. Frisch, M.J., Trucks, G.W., Schlegel, H.B., Scuseria, G.E., Robb, M.A., Cheeseman, J.R., Scalmani, G., Barone, V., Mennucci, B., Petersson, G.A., Nakatsuji, H., Caricato, M., Li, X., Hratchian, H.P., Izmaylov, A.F., et al., *Gaussian 09, Revision B.01*, Wallingford, Conn.: Gaussian, 2010.
27. Boese, A.D. and Martin, J.M.L., Development of density functionals for thermochemical kinetics, *J. Chem. Phys.*, 2004, vol. 121, no. 8, p. 3405.
28. Cossi, M., Scalmani, G., Rega, N., and Barone, V., New developments in the polarizable continuum model for quantum mechanical and classical calculations on molecules in solution, *J. Chem. Phys.*, 2002, vol. 117, no. 1, p. 43.
29. Schrodtt, A., Neubrand, A., and van Eldik, R., Fixation of CO_2 by zinc(II) chelates in alcoholic medium. X-ray structures of $\{[\text{Zn}(\text{cyclen})](3)(\mu(3)-\text{CO}_3)(\text{ClO}_4)(4)\}$ and $[\text{Zn}(\text{cyclen})\text{EtOH}](\text{ClO}_4)(2)$, *Inorg. Chem.*, 1997, vol. 36, no. 20, p. 4579.
30. Liptak, M.D. and Shields, G.C., Experimentation with different thermodynamic cycles used for pKa calculations on carboxylic acids using complete basis set and Gaussian-*n* models combined with CPCM continuum solvation methods, *Int. J. Quantum Chem.*, 2001, vol. 85, no. 6, pp. 727–741. doi 10.1002/qua.1703
31. Koziol, L., Valdez, C.A., Baker, S.E., Lau, E.Y., Floyd, W.C., Wong, S.E., Satcher, J.H., Lightstone, F.C., and Aines, R.D., Toward a small molecule, biomimetic carbonic anhydrase model: Theoretical and experimental investigations of a panel of zinc(II) azamacrocyclic catalysts, *Inorg. Chem.*, 2012, vol. 51, no. 12, p. 6803.
32. Mauksch, M., Brauer, M., Weston, J., and Anders, E., New insights into the mechanistic details of the carbonic anhydrase cycle as derived from the model system $(\text{NH}_3)(3)\text{Zn}(\text{OH})^+/\text{CO}_2$: How does the $\text{H}_2\text{O}/\text{HCO}(3)^-$ replacement step occur?, *Chembiochem*, 2001, vol. 2, no. 3, p. 190.
33. Huang, D.G., Makhlynets, O.V., Tan, L.L., Lee, S.C., Rybak-Akimova, E.V., and Holm, R.H., Kinetics and mechanistic analysis of an extremely rapid carbon dioxide fixation reaction, *Proc. Natl. Acad. Sci. U. S. A.*, 2011, vol. 108, no. 4, p. 1222.
34. Bhaduri, G.A. and Siller, L., Nickel nanoparticles catalyze reversible hydration of carbon dioxide for mineralization carbon capture and storage, *Catal. Sci. Technol.*, 2013, vol. 3, no. 5, p. 1234.
35. Steiner, H., Jonsson, B.H., and Lindskog, S., The catalytic mechanism of carbonic anhydrase. Hydrogen-isotope effects on the kinetic parameters of the human C isoenzyme, *Eur. J. Biochem.*, 1975, vol. 59, no. 1, pp. 253–259.
36. Silverman, D.N. and Lindskog, S., The catalytic mechanism of carbonic anhydrase: Implications of a rate-limiting protolysis of water, *Acc. Chem. Res.*, 1988, vol. 21, no. 1, pp. 30–36. doi 10.1021/ar00145a005
37. Simonsson, I., Jonsson, B.H., and Lindskog, S., A ^{13}C nuclear-magnetic-resonance study of $\text{CO}_2\text{-HCO}_3^-$ exchange catalyzed by human carbonic anhydrase C at chemical equilibrium, *Eur. J. Biochem.*, 1979, vol. 93, no. 2, pp. 409–417. doi 10.1111/j.1432-1033.1979.tb12837.x
38. Kimura, E., Shiota, T., Koike, T., Shiro, M., and Kodama, M., A zinc(II) complex of 1,5,9-triazacyclododecane ($[\text{12}]_{\text{aneN}_3}$) as a model for carbonic anhydrase, *J. Am. Chem. Soc.*, 1990, vol. 112, no. 15, pp. 5805–5811. doi 10.1021/ja00171a020
39. Krishnamurthy, V.M., Kaufman, G.K., Urbach, A.R., Gitlin, I., Gudiksen, K.L., Weibel, D.B., and Whitesides, G.M., Carbonic anhydrase as a model for biophysical and physical-organic studies of proteins and protein–ligand binding, *Chem. Rev.*, 2008, vol. 108, no. 3, pp. 946–1051. doi 10.1021/cr050262p
40. Kiefer, L.L., Paterno, S.A., and Fierke, C.A., Hydrogen-bond network in the metal-binding site of carbonic-anhydrase enhances zinc affinity and catalytic efficiency, *J. Am. Chem. Soc.*, 1995, vol. 117, no. 26, pp. 6831–6837. doi 10.1021/ja00131a004